

Run 03 | Fear of failure

Global (including India)

IV + controls

Target Fear of failure prevents starting a business; binary 1 vs. 0
Sample 1,334,667 respondents **Coverage** 89 countries; 2015–2022
Positive cases 583,346 (43.71%) **Case time** 36m 44s
Train / val. / test 907,575 / 160,159 / 266,933 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.6986). Highest F1: Random Forest (0.6447). For Random Forest, adding controls changed ROC-AUC by +0.0169 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.6986	0.6365	57.77	50.98	87.67	0.6447	5m 38s
XGBoost	0.6808	0.6099	56.82	50.35	86.71	0.6371	29s
H2O GBM	0.6774	0.6055	55.18	49.30	89.37	0.6355	2m 22s
CatBoost	0.6768	0.6060	56.61	50.21	86.50	0.6354	3m 29s
Logistic Regression	0.6249	0.5468	49.98	46.37	92.44	0.6176	1m 05s
Elastic Net	0.6249	0.5469	50.36	46.56	91.75	0.6177	4m 45s
Lasso	0.6248	0.5469	50.36	46.56	91.74	0.6177	4m 54s
Ridge	0.6247	0.5467	50.38	46.57	91.74	0.6178	8m 25s
Decision Tree	0.6118	0.5295	43.79	43.74	99.88	0.6083	30s
Naive Bayes	0.5892	0.5242	44.07	43.83	99.27	0.6081	4m 04s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Age	20,141.06	12.21
2	Household size	10,916.95	6.62
3	Occupational status	10,030.73	6.08
4	Education attainment	7,966.77	4.83
5	Household income	7,377.94	4.47
6	Entrepreneurial intention	5,736.96	3.48
7	Self-efficacy	5,527.95	3.35
8	Good career choice	4,283.53	2.60

Share is the percentage of total forest impurity importance across all 64 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	3,143.86	1.91
2	Economic quality	3,074.15	1.86
3	Social capital	2,764.90	1.68

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 4 batch, 29 September 2026; 03_obj4_global_iv_cv_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.